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Original Article

Developing machine learning models to predict primary graft dysfunction after lung transplantation

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ABSTRACT

Primary graft dysfunction (PGD) is the leading cause of morbidity and mortality in the first 30 days after lung transplantation. Risk factors for the development of PGD include donor and recipient characteristics, but how multiple variables interact to impact the development of PGD and how clinicians should consider these in making decisions about donor acceptance remain unclear. This was a single-center retrospective cohort study to develop and evaluate machine learning pipelines to predict the development of PGD grade 3 within the first 72 hours of transplantation using donor and recipient variables that are known at the time of donor offer acceptance. Among 576 bilateral lung recipients, 173 (30%) developed PGD grade 3. The cohort underwent a 75% to 25% train-test split, and lasso regression was used to identify 11 variables for model development. A K-nearest neighbor's model showing the best calibration and performance with relatively small confidence intervals was selected as the final predictive model with an area under the receiver operating characteristics curve of 0.65. Machine learning models can predict the risk for

Abbreviations: AI, artificial intelligence; AUROC, area under the receiver operating characteristics curve; AUPRC, area under the precision recall curve; BMI, body mass index; COPD, chronic obstructive pulmonary disease; CV, cross validation; ECMO, extracorporeal membrane oxygenation; EHR, electronic health record; F, female; FiO_2, fraction of inspired oxygen; HPO, hyperparamter optimization; ICU, intensive care unit; ILD, interstitial lung disease; IQR, interquartile range; KNN, K-nearest neighbors; LAS, lung allocation score; LR, logistic regression; ML, machine learning; PA, pulmonary artery; $PaCO_2$, partial pressure of arterial carbon dioxide; PaO_2, partial pressure of arterial oxygen; PEEP, positive end-expiratory pressure; PF, partial pressure of arterial oxygen: fraction of inspired oxygen; PGD, primary graft dysfunction; SHAP, shapley additive explanations; SVC, support vector classifier; UNOS, United Network for Organ Sharing; UNet, UNOS donor portal; XGB, extreme gradient boosting.

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development of PGD grade 3 based on data available at the time of donor offer acceptance. This may improve donor-recipient matching and donor utilization in the future.

1. Introduction

Lung transplantation is the ultimate treatment for patients with advanced lung disease, and approximately 4500 lung transplants are performed annually worldwide.[1] However, long-term outcomes after lung transplantation remain disappointing, and the median survival is 6.7 years.[1] Primary graft dysfunction (PGD) is a form of acute lung injury occurring in the first 72 hours after lung transplantation and is the leading cause of morbidity and mortality in the first 30 days after lung transplantation.[1] PGD is diagnosed and graded according to the presence of pulmonary edema on chest x-ray and impairment in gas exchange as assessed by partial pressure of arterial oxygen: fraction of inspired oxygen (PF) ratio.[2] Previous studies have shown that the incidence of PGD grade 3 within the first 72 hours after lung transplantation is 30%.[3] PGD has a significant impact on short-term outcomes after transplantation, including duration of mechanical ventilation, hospital length of stay, and survival.[4] Moreover, PGD has an adverse impact on long-term outcomes beyond the first year after transplantation.[5] We have previously shown that PGD is an independent risk factor for the development of bronchiolitis obliterans syndrome.[6,7] Thus, PGD adversely affects short- and long-term outcomes after lung transplantation, and strategies that mitigate the risk of PGD are needed to improve outcomes.

The pathogenesis of PGD is multifactorial, and it is believed to be the end result of multiple injuries that begin with donor brain death, ventilator-induced lung injury, aspiration, followed by cold ischemia after organ recovery, and reperfusion at the time of transplantation.[8] This results in noncardiogenic pulmonary edema and impaired lung function. Certain donor and recipient characteristics are associated with an increased risk of PGD.[3] However, some donor characteristics may be acceptable in a recipient who is considered at low risk for PGD, whereas a different combination of donor and recipient characteristics may have an additive or synergistic effect that becomes prohibitive.[9] How these variables interact in a given donor-recipient pair to impact the development of PGD is unclear. More importantly, how clinicians should consider these variables in making decisions about donor organ acceptance for a given recipient is also unclear. Currently, clinicians make predictions about the potential risk of PGD based on donor organ function (eg, PF ratio and imaging studies) and recipient characteristics to decide about organ acceptance, but these predictions are often imprecise and contribute to underutilization of suitable donors for certain recipients. Indeed, multiple studies have demonstrated good posttransplant outcomes without an excessive incidence of severe PGD using extended criteria donors who had PF ratios $<$ 300 mmHg or some other risk factor for PGD (eg, advanced age and smoking history).[10,11] Nevertheless, clinicians remain hesitant to accept donor lungs that are perceived to be of lower quality, and reliable predictive models are needed to expand the donor pool. Ex vivo lung perfusion has been increasingly used over the past 10 years to improve donor utilization.[12] However, this technology is expensive and labor- and time-intensive, and access to ex vivo lung perfusion at many centers across the world is limited. Thus, we sought to develop a machine learning (ML) model that could be used as an adjunctive donor assessment tool that predicts the development of PGD using preoperative donor and recipient variables known at the time of donor organ evaluation.

2. Materials and methods

2.1. Study population

All adult ($\geq$18 years of age) recipients of bilateral lung transplants at our center between January 1, 2010, and December 31, 2019, who consented to participate in our Lung Transplant Registry were included in the study. Patients were excluded if they underwent combined organ (eg, lung-liver and heart-lung) or single lung transplants. The final cohort consisted of 576 patients. The study protocol was approved by the Washington University Human Research Protection Office institutional review board (IRB #202012155).

2.2. Data collection

For all included recipients, data were collected and aggregated from the United Network for Organ Sharing (UNOS), the UNOS donor portal (UNet), and local electronic health record (EHR) system (Epic). Data from UNet and the local EHR were extracted by research coordinators and saved into a research electronic data capture database.[13] Data from research electronic data capture and UNOS were merged for model development and creation. Extracted data included pretransplant variables from both the donor (demographics, laboratory and vital sign values, ventilator parameters, mechanism of death, distance to transplant center, medications, etc.) and recipient (demographics, laboratory and vital sign values, location pretransplant, lung allocation score [LAS], respiratory support method pretransplant, performance status, right heart catheterization values, etc.).

2.3. Primary outcome

The primary outcome was PGD grade 3 at any time within 72 hours of transplantation, according to International Society for Heart and Lung Transplant criteria.[1] PGD was graded by 2 board-certified pulmonologists (A.M., R.H.). To be considered for PGD, chest x-ray reports had to include a predefined set of compatible phrases (Supplementary Table 1) as read by a board-certified radiologist.

2.4. Model development

The population was divided into a 75% to 25% train-test split. Categorical data were converted into binary features, after which 1 category was removed. The train and test data were scaled using sklearn.preprocessing. The train and test data were scaled using the sklearn.preprocessing.MinMaxScaler (feature range = (0,1)) fit to the train data and applied to the test data. Missing data were median imputed for continuous data and mode imputed for binary data. A lasso regression was performed to reduce the number of clinical features used for prediction, using Akaike's information criterion to select the optimal regularization parameter, alpha. Although this model was developed for predictive rather than explanatory analyses, the resultant features identified using lasso regression were evaluated for collinearity using Pearson's correlation coefficients. Figure 1 illustrates the modeling process. Using the reduced feature set as input, 4 class prediction algorithms—logistic regression (LR), K-nearest neighbors (KNN), extreme gradient boosting (XGB), and support vector classifier (SVC) with linear kernel were trained and tested. Hyperparameters were optimized using 5-fold cross validation with random grid search set to optimize log loss (Supplementary Table 2). Models were evaluated using area under the receiver operating characteristic curve (AUROC), the area under the precision recall curve (AUPRC), and F1 score. To further assess model consistency, we performed 50 bootstrapping iterations to generate median and interquartile ranges (IQRs) for each metric based on predictions on the test data after fitting the model to a resampled data set of the same size as the training data, derived by sampling the training data with replacement. Model calibration was assessed before (precalibrated) and after sigmoid calibration (calibrated) by calculating the Cox intercepts and slope and plotting calibration curves. This model was developed in accordance with the TRIPOD statement (Supplementary Fig. 1). Missing data are shown in Supplementary Table 3. All data cleaning and model development were performed using scikit-learn (version 1.0.2) in Python 3.8.5. Feature importance was assessed using the shapley additive explanations package (version 0.39.0). Figures were made using draw.io.

2.5. Statistical analysis

Results are summarized using frequencies and proportions for categorical data or medians and IQR for continuous data. Between-group comparisons were performed using the 2-sided Mann-Whitney U test for continuous data and the 2-sided Fisher exact test for categorical data. A P value <0.05 was considered statistically significant. Analyses and figure generation were performed using R version 4.1.2 (R Foundation for Statistical Computing) using RStudio (RStudio, PBC).

2.6. Website development

The model with the best balance between AUROC, AUPRC, and calibration was selected for website implementation. The model was embedded within a HIPAA-secure Google Cloud environment and is accessible through permissioned access at https://pgdcalc.wustl.edu.

3. Results

3.1. Study population

Among 580 bilateral lung transplant recipients, 576 were included in the Lung Transplant Registry and available for analysis. PGD grade 3 at any time in the first 72 hours after lung transplantation occurred in 173 (30%) patients (Fig. 2). Compared to patients who did not develop PGD grade 3 (no PGD, n = 403), patients who developed PGD were more likely to be female (85/173 [49%] vs 152/403 [38%], $P<.05$); be Black (20/173 [12%] vs 17/403 [4%], $P<.005$); have a higher LAS (median [IQR]: 40.6 [35.9-54.4] vs 37.3 [33.5-45.4], $P<.001$); have a primary diagnosis of interstitial lung disease (97/173 [56%] vs 173/403 [43%], $P=.005$); have been in the intensive care unit (ICU) pretransplant (22/173 [18%] vs 14/403 [5%], $P<.05$); require mechanical ventilation pretransplant (11/173 [13%] vs 7/403 [3%], P

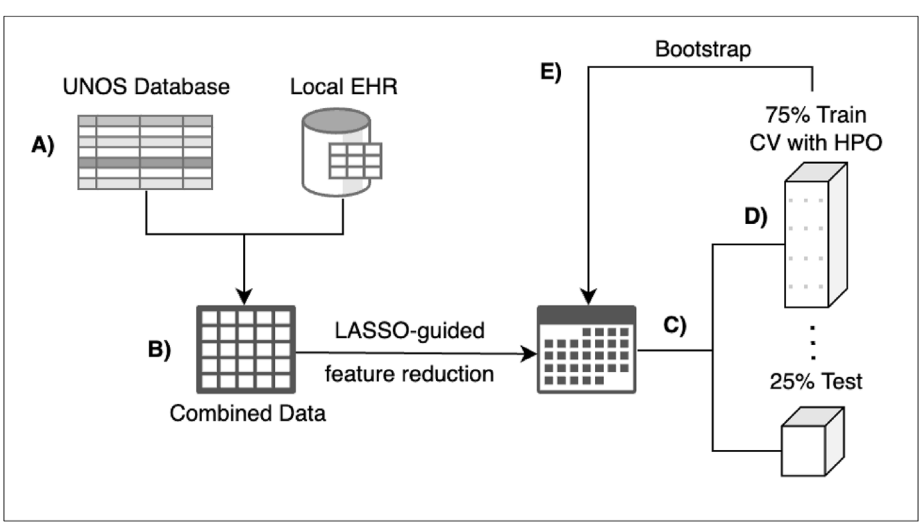

Figure 1. Modeling process. (A) Patients in the Lung Transplant Registry had their data extracted from the United Network for Organ Sharing (UNOS) database and the local EHR. (B) These data were combined and fed into a lasso regression to identify the most important clinical features required for predicting primary graft dysfunction. (C) These data were then split into a training and test data set. (D) Five-fold cross validation was performed on the training data and used for hyperparameter optimization. (E) This process was performed 50 times in a bootstrap analysis to generate estimates of error. This process (C through E) was repeated for each type of machine learning model. CV, cross validation; EHR, electronic health record; HPO, hyperparamter optimization.

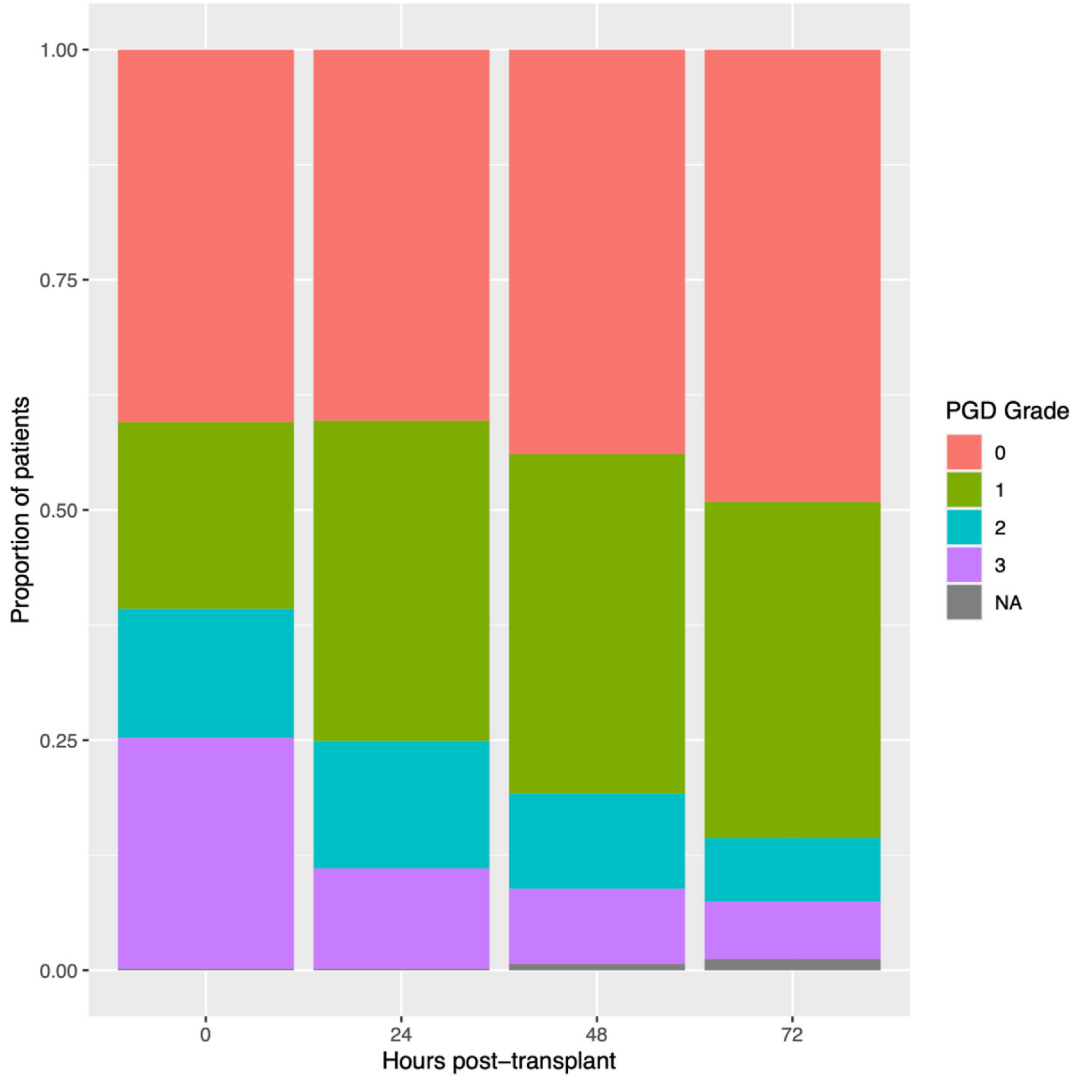

Figure 2. Distribution of primary graft dysfunction (PGD) grade at 0, 24, 48, and 72 hours posttransplant. NA, not applicable.

< .01); and have a donor who experienced chest trauma (26/173 [15%] vs 45/403 [11%], $P < .05$). They were also less likely to be White (149/173 [86%] vs 383/403 [95%], $P < .001$) and have a primary diagnosis of chronic obstructive pulmonary disease (32/173 [19%] vs 132/403 [33%], $P < .001$) (Table 1). No significant differences between those who developed PGD and those who did not were observed for donor or recipient age and sex; donor body mass index, cause of death, cigarette use, and last recorded PF ratio; or recipient mean pulmonary artery pressure and underlying diagnoses of cystic fibrosis, pulmonary hypertension, or sarcoidosis.

3.2. Model development

Following feature engineering, the final data set encompassed 100 features (49 from UNOS and UNet and 51 from the local EHR). Lasso regression reduced the feature set to 11 pretransplant variables that were subsequently used for model development (Table 2). There was no strong collinearity between any 2 features, although there was moderate colinearity between LAS and recipient ICU admission pretransplant (correlation coefficient = 0.6), Supplementary Table 4. As shown in Table 3, which compares the best-performing model for each algorithm selected during hyperparameter optimization (Supplementary Table 2), before calibration, XGB delivered the model with the highest AUROC (0.72), AUPRC (0.6), and F1 score (0.67). However, the bootstrap analysis showed that SVC produced a higher median AUROC and high lability of the XGB and SVC models, as shown by the wider range of AUROC, AUPRC, and F1 scores compared to the LR and KNN models (Fig. 3A-C). After calibration of the gridsearch-selected best models, XGB again produced the highest AUROC (0.69), AUPRC (0.47), and F1 (0.65) (Table 3). This XGB model also produced the highest median AUROC, AUPRC, and F1 from bootstrapping, but the variability in metric values produced by this analysis demonstrated that the XGB model's predictions were less stable to variations in the data on which it was trained (Fig. 3D-F).

Precalibration, the SVC model had the smallest intercept (−0.295) and a slope closest to 1 (0.642; Table 4; Supplementary Fig. 2). After calibration, the KNN model is the best calibrated, having the smallest intercept (0.075) and slope closest to 1 (1.152). Visual comparison of the calibration curves for the

Table 1
Baseline characteristics of donors and recipients.

Variable	Total (N = 576)	PGD (N = 173)	No PGD (N = 403)	*P* value
Donors				
Age (y), median (IQR)	33 (23-48)	35 (25-48)	32 (22-48)	.082
Sex, n (%)				
Male	345 (59.9%)	110 (63.58%)	235 (58.31%)	.266
Female	231 (40.1%)	63 (36.42%)	168 (41.69%)	
Missing	0 (0%)	0 (0%)	0 (0%)	
Race, n (%)				
White	424 (73.61%)	132 (76.3%)	292 (72.46%)	.355
Black	119 (20.66%)	33 (19.08%)	86 (21.34%)	.576
Other	33 (5.73%)	8 (4.62%)	25 (6.2%)	.206
Cause of death				
Cerebrovascular stroke	175 (30.38%)	52 (30.06%)	123 (30.52%)	1
Head trauma	237 (41.15%)	74 (42.77%)	163 (40.45%)	.644
Anoxia	140 (24.31%)	41 (23.7%)	99 (24.57%)	.916
Other	24 (4.17%)	6 (3.47%)	18 (4.47%)	.656
Cigarette use > 20 pack-y, n (%)	53 (9.2%)	19 (10.98%)	34 (8.44%)	.347
Cigarette use within past 6 mo, n (%)	87 (15.1%)	26 (15.03%)	61 (15.14%)	1
Chest trauma, n (%)	71 (12.33%)	26 (15.03%)	45 (11.17%)	.043
Last PaO_2:FiO_2 ratio median (IQR)	452 (392-506)	441 (374-496)	456 (400-509)	.054
Recipients				
Age, median (IQR)	59 (48-65)	57 (47-66)	60 (50-65)	.243
Sex, n (%)				
Male	338 (58.68%)	88 (50.87%)	250 (62.03%)	.019
Female	237 (41.15%)	85 (49.13%)	152 (37.72%)	
Missing	1 (0.17%)	0 (0%)	1 (0.25%)	
Race, n (%)				
White	532 (92.36%)	149 (86.13%)	383 (95.04%)	.000
Black	37 (6.42%)	20 (11.56%)	17 (4.22%)	.002
Other	7 (1.22%)	4 (2.31%)	3 (0.74%)	.559
BMI (kg/m^2) median (IQR)	24.5 (20.6-27.9)	25 (21.1-27.8)	24.3 (20.2-27.9)	.342
Underlying diagnosis				
Interstitial lung disease	270 (46.88%)	97 (56.07%)	173 (42.93%)	.005
COPD	164 (28.47%)	32 (18.5%)	132 (32.75%)	<.001
Cystic fibrosis	93 (16.15%)	26 (15.03%)	67 (16.63%)	.711
PH	14 (2.43%)	6 (3.47%)	8 (1.99%)	.375
Sarcoidosis	8 (1.39%)	2 (1.16%)	6 (1.49%)	1
Other	27 (4.69%)	10 (5.78%)	17 (4.22%)	.399
Lung Allocation Score median (IQR)	38.3 (34.1-46.7)	40.6 (35.9-54.4)	37.3 (33.5-45.4)	<.001
Hospitalized pretransplant, n (%)	167 (28.99%)	59 (34.1%)	108 (26.8%)	.088

(*continued on next page*)

Table 1 (*continued*)

Variable	Total (N = 576)	PGD (N = 173)	No PGD (N = 403)	*P* value
In the ICU pretransplant, n (%)	36 (6.25%)	22 (12.72%)	14 (3.47%)	<.001
Requiring mechanical ventilation pretransplant, n (%)	18 (3.12%)	11 (6.36%)	7 (1.74%)	.007
Requiring ECMO pretransplant, n (%)	8 (1.39%)	5 (2.89%)	3 (0.74%)	.057
Pulmonary hypertension, n (%)	14 (2.43%)	6 (3.47%)	8 (1.99%)	.375
Mean PA pressure median (IQR)	27 (22-32)	28 (22-35)	26 (22-32)	.188

BMI, body mass index; COPD, chronic obstructive pulmonary disease; ECMO, extracorporeal membrane oxygenation; FiO_2, fraction of inspired oxygen; ICU, intensive care unit; IQR, interquartile range; PA, pulmonary artery; PaO_2, partial pressure of arterial oxygen; PGD, primary graft dysfunction; PH, pulmonary hypertension.

Table 2
Lasso selected features.

Feature name	Description
LAS	Last LAS before transplantation
Donor $PaCO_2$	Last recorded $PaCO_2$ for the donor
White recipient	Recipient race: White
COPD	Recipient underlying diagnosis: COPD
ILD	Recipient underlying diagnosis: ILD
Recipient sex	Recipient sex
Recipient ICU admission pretransplant	Recipient in the ICU immediately before transplantation
Donor chest trauma	Donor suffered chest trauma before retrieval
Donor PEEP	Last recorded PEEP for the donor
Female donor:male recipient	Female donor sex and male recipient sex
Other donor race	Donor race is not White or Black

COPD, chronic obstructive pulmonary disease; ICU, intensive care unit; ILD, interstitial lung disease; LAS, Lung Allocation Score; $PaCO_2$, partial pressure of arterial carbon dioxide; PEEP, positive end-expiratory pressure.

Table 3
Comparison of performance metrics from hyperparameter optimization.

Model	Calibration status	5-fold CV best score		
		AUROC	AUPRC	F1
LR	Precalibrated	0.63	0.38	0.61
KNN	Precalibrated	0.64	0.43	0.63
	Calibrated	0.65	0.45	0.62
XGB	Precalibrated	0.72	0.6	0.67
	Calibrated	0.69	0.47	0.65
SVC	Precalibrated	0.63	0.38	0.57
	Calibrated	0.55	0.34	0.51

AUROC, area under the receiver operating characteristics curve; AUPRC, area under the precision recall curve; CV, cross validation; KNN, K-nearest neighbors; LR, logistic regression; SVC, support vector classifier; XGB, extreme gradient boosting.

precalibrated and calibrated KNN models reveal that despite worse Cox intercept and slope values, the precalibrated KNN model also appears well-calibrated (Fig. 4 and Supplementary Fig. 2). Furthermore, the precalibrated KNN model generates a wider range of predictions indicating that it is less conservative in its predictions than the calibrated KNN model (Fig. 4). As the KNN model had the best calibration with acceptable performance and relatively small confidence intervals, it was selected for further evaluation, and the relative feature importance of the included variables can be seen in Figure 5. Because the lung allocation system changed in March 2023, we repeated the analyses excluding LAS (Supplementary Tables 5, 6 and Supplementary Figs. 4-6). To make this predictive model readily available, it was deployed in a HIPPA-secure Google cloud instance, which can be accessed at https://pgdcalc.wustl.edu. Users can enter donor and recipient features and view the estimated risk for PGD development based on presurgical clinical variables (Fig. 6).

4. Discussion

ML is a branch of artificial intelligence (AI) that is capable of ingesting large volumes of data to facilitate the discovery of novel insights. Commonly, it is used for classification tasks and outcome probability assessments. Over the past 10 years, it has been increasingly applied to health care, with expanding applications within organ transplantation, often evaluating its utility in predicting survival after heart, liver, and kidney transplantation.[14-20] More recent investigations have used AI and ML to predict stable tacrolimus doses and optimal donor-recipient matching.[15,21,22] To date, however, AI and ML have not been thoroughly evaluated in terms of their ability to predict outcomes after lung transplantation. In this retrospective analysis, donor and recipient data that would be known at the time of donor organ acceptance were used to create a predictive model for PGD grade 3. Using these data, a series of AI and ML models were trained and tested in a rigorous fashion, which resulted in models with relatively high precision and good calibration. To the best of our knowledge, this is the first application of AI/ML to predict the risk of PGD. It is also an initial attempt to move beyond identifying individual clinical risk factors associated with PGD. In the future, accurate and precise models of predicting PGD may ameliorate the severity of PGD and improve donor and recipient matching and donor utilization by identifying suitable recipients for donors deemed to be at unacceptably high risk for severe PGD.

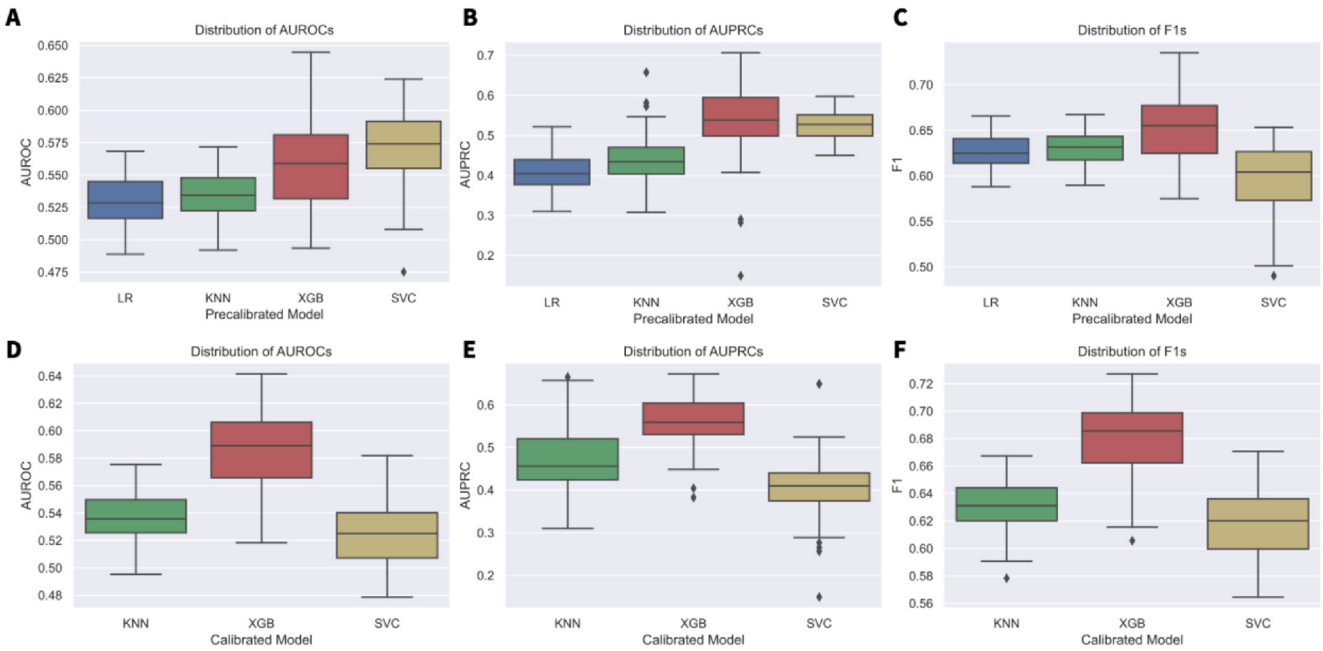

Figure 3. Consistency of model performance. Distribution of area under the receiver operating characteristics curve (AUROC) (A, D), area under the precision recall curve (AUPRC) (B, E), and F1 (C, F) for the precalibrated (A, B) and calibrated (C, D) models was obtained using bootstrapping. Box and whisker plots indicate median and interquartile range (IQR) (box), with the whiskers indicating 1.5 IQRs of the lower and upper quartiles. KNN, K-nearest neighbors; LR, logistic regression; SVC, support vector classifier; XGB, extreme gradient boosting.

Table 4
Calibration summary.

Model	Precalibrated		Calibrated	
	Intercept	Slope	Intercept	Slope
LR	−0.365	0.586	NA	NA
KNN	−0.349	0.439	0.075	1.152
XGB	0.885	1.994	−0.764	1.318
SVC	−0.295	0.642	−0.836	0.464

Intercept: 0 = good calibration, >0 = average underestimation, <0 = average overestimation.
Slope: 1 = perfect calibration or overfit, >1 = underestimation of high risk and overestimation of low risk, <1 = underestimation of low risk and overestimation of high risk.
KNN, K-nearest neighbors; LR, logistic regression; NA, not applicable; SVC, support vector classifier; XGB, extreme gradient boosting.

In the population studied, PGD grade 3 occurred in 173 (30%) patients, which is consistent with previous studies.[3] In addition, unadjusted analyses recapitulated known risk factors for PGD, such as recipient female sex, LAS, interstitial lung disease as the underlying diagnosis leading to transplant, recipient admission to the ICU before transplantation, and requiring invasive mechanical ventilation before transplant.[3,23-25] Together, the incidence of PGD and the demonstration of known risk factors associated with PGD suggest that our population was appropriate for the prediction task.

After merging UNOS, UNet, and local EHR data, more than 100 features for model prediction were created. In order to narrow the feature space, we used a standard lasso regression to select only those features that were predictive of the outcome of interest. Ultimately, 11 features were selected, including some of the features associated with the outcome in univariate analyses. Interestingly, additional features about the donor lungs emerged, including the last donor's partial pressure of carbon dioxide ($PaCO_2$) and positive end-expiratory pressure. The feature importance plots for the selected model suggest that higher $PaCO_2$ is associated with a higher probability of PGD. Although it is difficult to explain this association, $PaCO_2$ may be a surrogate marker for subclinical lung injury that requires permissive hypercapnia to minimize airway pressures. Similarly, higher positive end-expiratory pressure values after donor stabilization suggest some degree of lung injury. Interestingly, some previously recognized risk factors for PGD were not chosen as important features in this model. It is possible that this population is uniquely different; indeed, our cohort included a very small number of patients who required extracorporeal membrane oxygenation support before transplant or had pulmonary hypertension as the indication for transplant. In addition, it is likely that our novel model allows a relatively large number of clinical variables to interact to identify new combinations of clinical data that are important for outcome prediction.

After model training, validation, and testing, a bootstrap analysis was performed to estimate the general performance of the model when different groups of patients were used for training, validation, and testing. Although more computationally expensive, this provides a better estimate of expected model performance, which is particularly important when the total population is relatively small, as it is here. That said, the achieved measures of model performance, namely AUROC, remained modest (between 0.5 and 0.6) despite the large feature space analyzed. This less than ideal AUROC likely arises from multiple etiologies, including the limited sample size, the fact that hyperparameters were

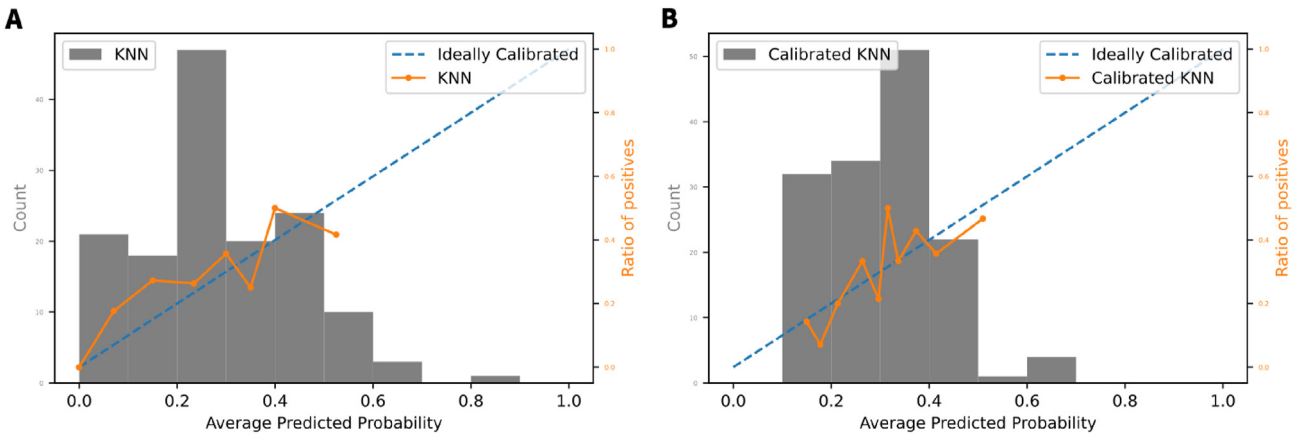

Figure 4. Calibration of K-nearest neighbors (KNN) models before (A) and after (B) sigmoid calibration. The gray histograms represent the number of samples binned by predicted probability in intervals of 0.1. The orange line plots the actual ratio of positives against the average predicted probability, where each data point represents an equal number of samples.

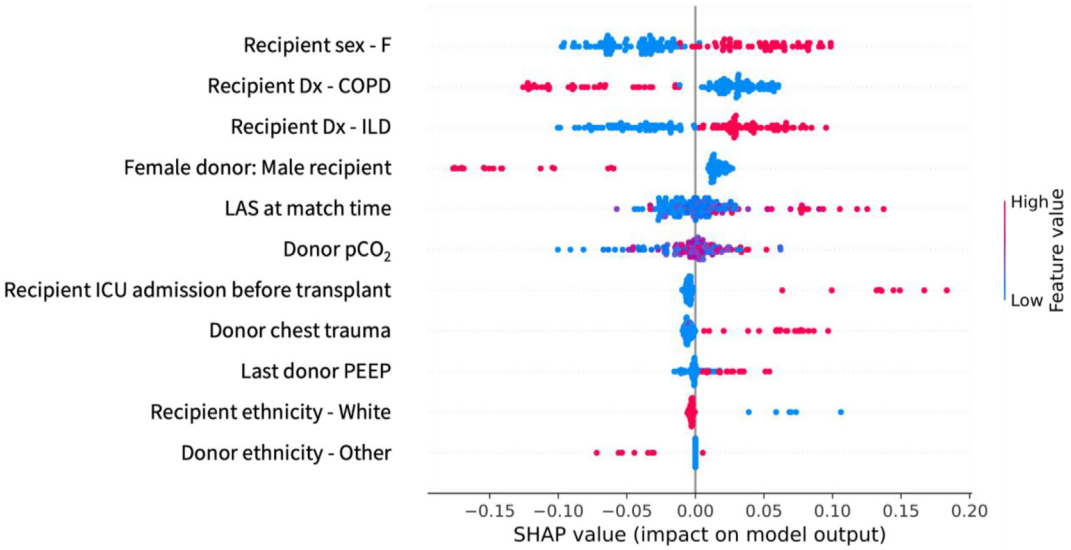

Figure 5. Shapley additive explanations (SHAP) plot of feature importance for the K-nearest neighbor's model. Features are arranged from top to bottom in order of decreasing impact on the model's prediction of primary graft dysfunction (PGD) risk. The x-axis indicates the magnitude and direction of the feature's impact on the model's prediction (positive = PGD, negative = no PGD). Each point represents a patient sample, with the color indicating the feature value (red = high value, binary feature present [1]; blue = low value, binary feature absent [0], male sex is represented by a 0). COPD, chronic obstructive pulmonary disease; F, female; ICU, intensive care unit; ILD, interstitial lung disease; LAS, lung allocation score; $PaCO_2$, partial pressure of carbon dioxide; PEEP, positive end-expiratory pressure.

optimized for log loss and not AUROC to promote optimal model calibration, and inherent difficulties predicting PGD using only variables known at the time of donor acceptance. Recent studies using LR achieved similar results in patients with pulmonary arterial hypertension who underwent lung transplantation. In this analysis, the best AUROC was 0.61, which is similar to the best cross validation performance seen in this study (Table 3). Likewise, a recent study used clinical variables to group patients into mortality risk groups and plotted them using Kaplan-Meier curves to generate risk of 1-, 5-, and 10-year mortality. In this study, the investigators achieved an impressive c-statistic of 0.72 for 10-year survival but used surgical and transplant-related data points (eg, ischemic time and need for extracorporeal membrane oxygenation support).[26] Because surgical and transplant outcomes are not known at the time of donor acceptance or the start of transplant surgery, we did not include these in our predictive model.

Despite the modest c-statistics obtained, the AUPRC values for the models were fairly high, ranging between 0.4 and 0.7, suggesting a good positive predictive value. Not surprisingly, LR had an AUROC that performed similarly to the other predictive models; however, the precision was lower than that seen with more modern ML approaches, such as XGBoost and SVC-based models. Similarly, XGBoost tends to outperform other ML approaches using discrete data, but in this instance, SVC appeared to have lower degrees of variance and better calibration. This result is likely due to the theoretical approaches behind the

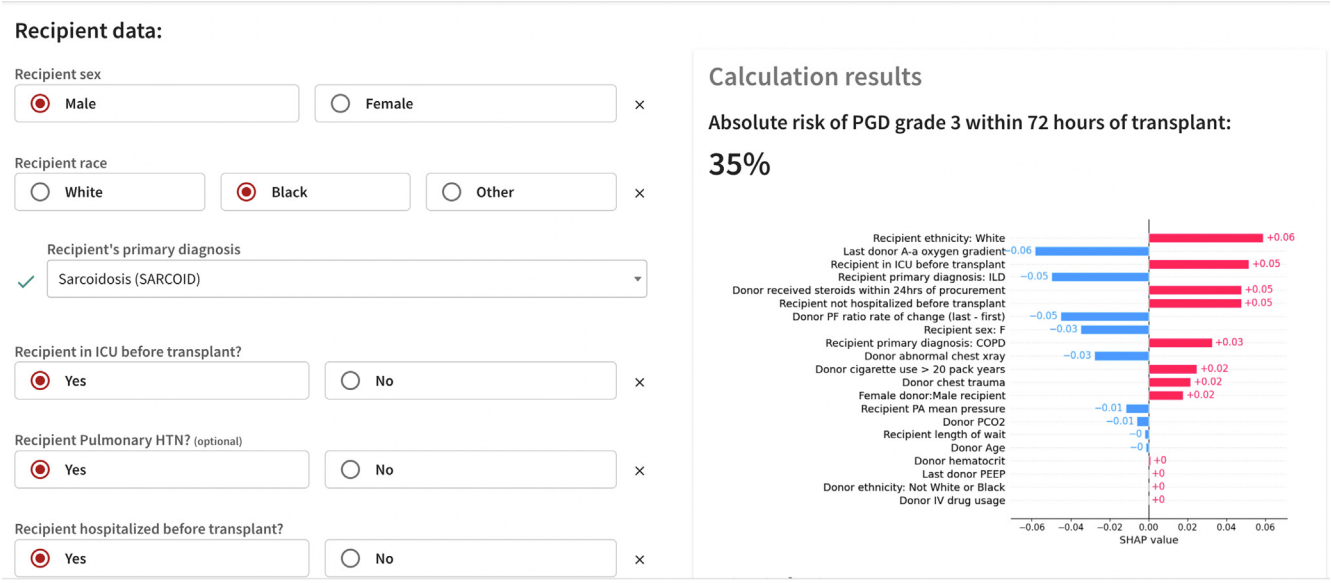

Figure 6. Screenshot of PGDcalc website, showing included variables and estimated risk of primary graft dysfunction (PGD) for a theoretical donor-recipient pair. Website available at www.PGDCalc.wustl.edu. COPD, chronic obstructive pulmonary disease; F, female; FiO_2, fraction of inspired oxygen; HTN, hypertension; ICU, intensive care unit; ILD, interstitial lung disease; LAS, lung allocation score; PA, pulmonary artery; $PaCO_2$, partial pressure of carbon dioxide; PEEP, positive end-expiratory pressure; SHAP, shapley additive explanations.

individual algorithms that allow SVC to work better for smaller populations than XGBoost. Interestingly, although the XGBoost model had the highest median AUROC, it had the widest confidence intervals, suggesting that with larger populations, performance could be further enhanced.

Overall, model calibration was fairly good but deviated at extremes of the predictive range. This occurs due to the small number of patients with very high or very low predicted risk. Nevertheless, the precalibrated KNN had the second-best slope and intercept values (Table 3), and overall best fit line that encompassed the extremes of the predicted probability. As a result, the precalibrated KNN was chosen as the optimal model for use on the publicly available PGD prediction website: https://pgdcalc.wustl.edu.

Over time, the model can be refined further with a larger sample size, and ensemble methods can be crafted to boost performance. Similarly, the model could be further augmented by incorporating imaging and biomarker data. This study, however, has inherent limitations as a single-center experience where all donors assessed were accepted for transplantation. Thus, our data and model are not designed for assessing the suitability of donor lungs. In addition, intraoperative and surgical parameters could change the predicted probability, and the model could fall out of calibration as the field evolves. We did not include intraoperative variables in the model because these are largely unknown at the time of donor evaluation. Finally, this model included LAS, but the composite allocation score will need to be considered as a potential feature in the future. The model requires external and prospective validation. Additional future work includes incorporating donor computed tomography scan images into the model. Indeed, the model lends itself to incorporating donor and recipient biomarkers and genetic data to better predict the development of PGD in the future.

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Disclosure

The authors of this manuscript have conflicts of interest to disclose as described by the American Journal of Transplantation. R.R.H. serves on the Clinical Events Adjudication Committee for TransMedics, Inc. D.K. is on the Scientific Advisory Board of Sana Biotechnology. Other authors have no conflicts of interest to disclose.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of interests

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ramsey Hachem, Andrew Michelson, Inez Oh, Aditi Gupta, Philip Payne reports financial support was provided by Mid-America Transplant Foundation. Ramsey Hachem reports a relationship with TransMedics Inc that includes: consulting or

advisory. Daniel Kreisel is on the Scientific Advisory Board of Sana Biotechnology

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajt.2023.07.008.

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